

SHOWTELLARENA: Evaluating Whether Agents Learn from Human Demonstrations

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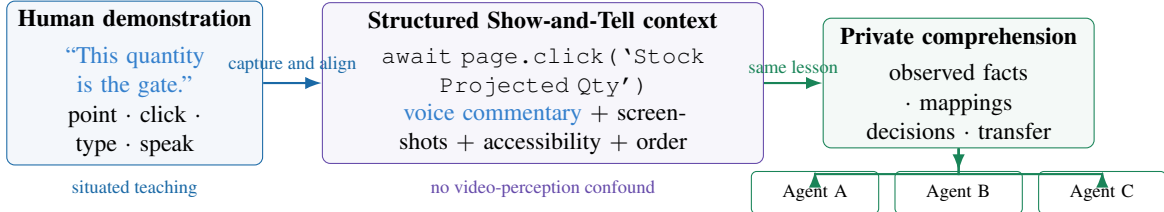


Figure 1: From showing to knowing. A person points into a software scene while explaining what matters. Brackett compiles pointing, actions, spoken reasoning, screen state, and accessibility semantics into one temporally coherent lesson. SHOWTELLARENA gives the same structured lesson to different agents and privately quizzes what each learned. The canonical context is not a video: it is an inspectable, replayable account of what was shown and told.

Abstract

People rarely give a colleague rich context by writing a complete specification. They open the relevant material, point to the evidence, speak while it is visible, and answer questions. This may teach a procedure, explain one unusual case, communicate a judgment, or make a relationship explicit. Yet agent benchmarks usually begin after this teaching step: the context or goal is already written down, and the model is evaluated on execution or question answering. We present SHOWTELLARENA, a benchmark that measures whether an agent can learn situated knowledge from one human demonstration. Each lesson begins with a person interacting with a resettable software environment while explaining what matters. Brackett turns this interaction into a structured context: a Playwright-style action trace with aligned voice commentary, screen states, screenshots, and accessibility semantics. The learner receives this representation rather than a video, separating comprehension from long-video perception. A private quiz asks what was shown, what happened in what order, how entities mapped across applications, why a decision was made, whether its rule transfers, and what the lesson did not show. The working collection contains 63 scenarios and 842 questions across enterprise and web settings; its human-demonstration core is expanding toward the frozen release. We compare frontier agents under the same context and report accuracy, paired disagreement, performance by question family, abstention, and ablations over both context and demonstration. Complementary takes repeat the same lesson twice, then change one sealed fact in a third take; matched questions test whether answers remain stable when the lesson stays the same and follow the evidence when it changes. SHOWTELLARENA turns “show the agent once” into a reproducible test of what the agent learned.

1 Introduction

People teach by establishing shared attention. They open the relevant screen, point to an object or value, speak while the evidence is present, and answer questions. A buyer may explain why one request should be escalated; an analyst may point to the part of a chart that makes a result suspicious; a teammate may show why two records that look different refer to the same case. The actions matter, but they are not the lesson by themselves. The lesson also lives in what the person notices, which values they compare, how they connect evidence, and what they say while it is on screen.

This interaction is becoming an interface to models. Brackett’s SHOW AND TELL, Claude’s Teach experience, and screen-recording agents invite a person to show and speak instead of compressing the lesson into a prompt. A demonstration may communicate a repeatable procedure, but it may just as naturally explain one case, artifact, exception, or judgment that resists complete written specification.

Current agent benchmarks ask a different question. WebArena Zhou et al. (2024), VisualWebArena Koh et al. (2024), WorkArena Drouin et al. (2024), and OSWorld Xie et al. (2024) begin with a textual goal and measure whether an agent reaches the desired state. These benchmarks are essential for execution, but the specification has already been supplied. They cannot tell us whether an agent can acquire that specification from situated human teaching.

Demonstration benchmarks move closer, but typically represent the lesson as video, an unnarrated screen recording, or an action sequence Jang et al. (2025); Wornow et al. (2024); Deng et al. (2023). This entangles two questions: did the learner understand the work, or did it merely fail to perceive a long recording? It also discards useful structure that modern software exposes: exact actions, element identity, tab continuity, accessibility semantics, and the correspondence between a spoken explanation and the state being discussed.

SHOWTELLARENA evaluates the missing learning step. A human explains or performs something once in a deterministic software environment and speaks naturally while interacting with it. Brackett turns the recording into a temporally coherent lesson: executable-style actions, aligned commentary, screenshots, and accessibility state. Different agents receive the same lesson. They are then asked questions whose answers depend on attending to what was shown and told.

The benchmark separates four kinds of learning. *Observation* questions test details that appeared in the episode. *Relations and process* questions test ordering, state changes, and cross-system mappings. *Decision* questions test which evidence controlled a judgment. *Transfer and limits* questions test whether a learner can apply an established rule to a new case, or refuse a conclusion the lesson never supported. A single aggregate score is insufficient: an agent may remember identifiers while missing the decision rule, or infer a plausible policy while overlooking the demonstrated facts.

Software work is the benchmark’s first testbed, not the boundary of the capability. A workflow is one kind of lesson. The more general question is whether a model can form grounded understanding from a person speaking, pointing, showing, and responding in context.

Our contributions are:

1. **A benchmark for learning from demonstration.** SHOWTELLARENA measures comprehension after one situated lesson, rather than task completion from a written specification.
2. **A structured, non-video lesson representation.** Human voice, pointing through action, screenshots, accessibility semantics, and temporal continuity are compiled into an inspectable context shared across agents.
3. **A question set grounded in demonstrated work.** Questions cover facts, mappings, process, judgment, transfer, and what the lesson did not establish, with exact answers or explicit grading rubrics.
4. **A paired evaluation protocol.** Every learner receives the same lesson and questions. Related takes test invariance when the taught facts stay fixed and sensitivity when one sealed fact changes; results retain question-level outcomes, ablations, and discordance instead of collapsing immediately to one mean.

The benchmark and evaluation harness are released as an open dataset. Brackett’s context-construction stack remains a production system; evaluation begins from the frozen lesson bundles. Every agent receives the same versioned context, questions, and grading contract. A published score can therefore be reproduced without access to the collection stack.

2 Related Work

Web and computer-use benchmarks. WebArena [Zhou et al. \(2024\)](#) introduced self-hosted web applications with functional evaluation. VisualWebArena [Koh et al. \(2024\)](#) added visually grounded tasks; WorkArena [Drouin et al. \(2024\)](#) and AssistantBench [Yoran et al. \(2024\)](#) expanded

toward enterprise and long-horizon knowledge work; OS-World [Xie et al. \(2024\)](#) moved from the browser to general desktop interaction. These benchmarks measure whether an agent can execute a goal it is given. SHOWTELLARENA instead measures whether the agent can recover the goal, procedure, and relevant judgment from a demonstration.

Demonstrations and workflow induction.

Mind2Web [Deng et al. \(2023\)](#) provides human action sequences as supervised targets for web navigation. MOTIF [Burns et al. \(2022\)](#) pairs high-level mobile commands with demonstrations containing screens, view hierarchies, and actions, then evaluates feasibility and execution. Agent Workflow Memory [Wang et al. \(2024\)](#) and Learn-by-Interact [Su et al. \(2025\)](#) show that reusable workflows can be induced from prior trajectories or synthetic interaction. These works establish that experience can improve future action. Our unit is different: one human explicitly teaches something in a situated software lesson, and the learner is evaluated immediately on what that lesson established.

Static interface comprehension. WebSRC [Chen et al. \(2021\)](#) grounds questions in webpage text, HTML structure, layout, and screenshots; Screen2Words [Wang et al. \(2021\)](#) summarizes a mobile screen from pixels and UI structure; and ScreenQA [Hsiao et al. \(2025\)](#) supplies answer text and box-level evidence for questions over complete mobile screenshots. These datasets establish that interface meaning cannot be reduced to flattened text or pixels alone. Their unit, however, is one static screen or page segment. SHOWTELLARENA asks what persists across a taught episode and includes temporal relations, spoken rationale, cross-application identity, rule transfer, and what the lesson did not show.

Recorded process understanding.

WONDER-BREAD [Wornow et al. \(2024\)](#) evaluates business-process documentation, knowledge transfer, and improvement from screen recordings. GUI-World [Chen et al. \(2025\)](#) studies question answering over GUI videos. VideoWebArena [Jang et al. \(2025\)](#) is the closest comparison: narrated web tutorials support factual-retention and downstream skill tasks, and the authors find that tutorials may even hurt task performance. SHOWTELLARENA shares the interest in tutorial comprehension, but removes video perception as the canonical bottleneck and preserves the native structure of software interaction.

Multimodal content as structured context.

MM-Vid [Lin et al. \(2023\)](#) showed that long multimodal recordings can be transformed into coherent textual scripts that support grounded reasoning beyond the input length of short-clip models. SHOWTELLARENA adopts the broader principle while specializing it to interactive software. Its representation retains action semantics, page continuity, aligned explanation, pixels, and accessibility state rather than flattening the episode into prose alone.

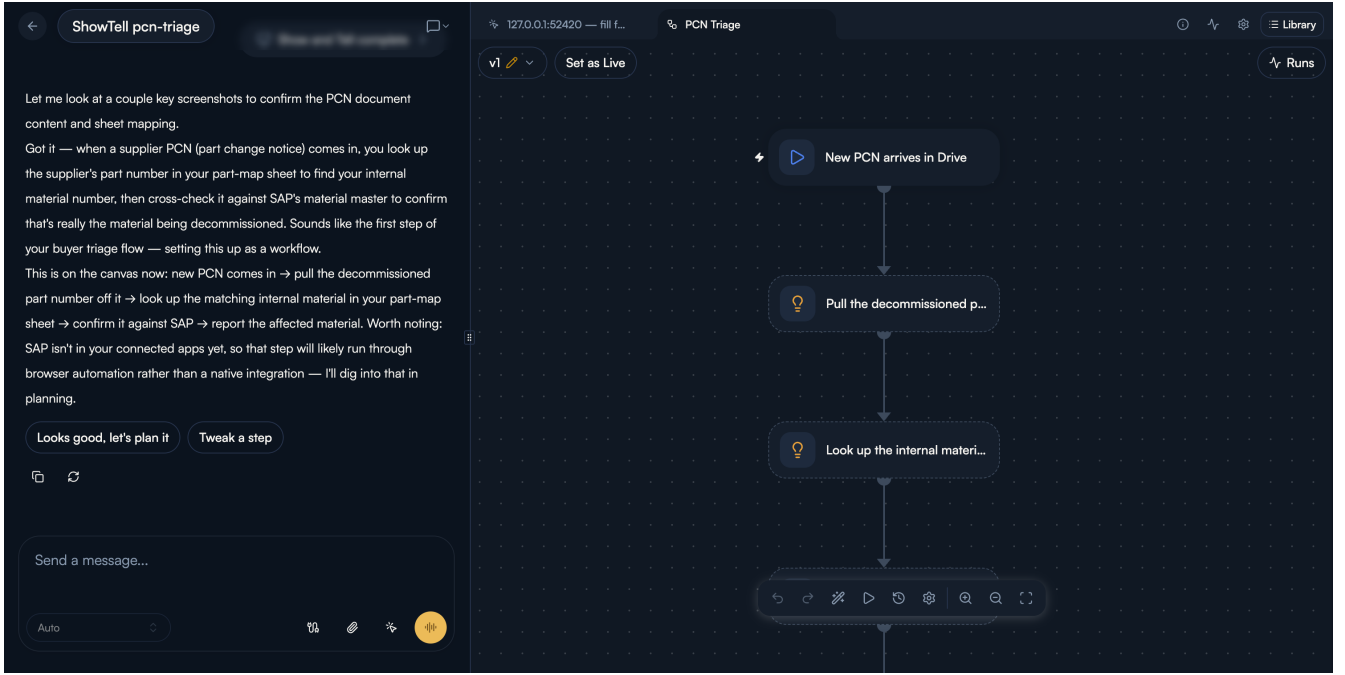


Figure 2: Show and Tell as a learning interface. The demonstration is not reduced to clicks. Spoken explanation, referenced screen state, application transitions, and observed values become a coherent account that an agent can inspect and reason over. The benchmark evaluates what downstream learners recover from that account.

	Human lesson	Spoken reasoning	Structured context	Comprehension	Same lesson across agents
WebArena	✗	✗	goal + environment	✗	✓
Mind2Web	action trace	✗	actions + DOM	next action	✓
WONDERBREAD	recording	✗	video	✓	✓
VideoWebArena	tutorial	narration	video	✓	✓
GUI-World	recording	limited	video	✓	✓
SHOWTELLARENA	✓	✓	trace + voice + screen + a11y	✓	✓

Table 1: Positioning. SHOWTELLARENA is designed around a human explicitly teaching through software and a common structured representation that lets us compare learners without turning the benchmark into a long-video perception test.

3 The SHOWTELLARENA Benchmark

SHOWTELLARENA is organized around a simple sequence: a person teaches, Brackett constructs the lesson context, an agent studies it, and a private evaluator asks what the agent learned. The environment, lesson, questions, and grading contract are frozen together so every learner encounters the same demonstrated world.

3.1 One situated human lesson

A demonstration takes place in the software context being explained. The teacher may stay with one artifact or move among an ERP, a spreadsheet, mail, chat, or another application. They interact normally and explain what they attend to: “I use the shortage quantity here, not the value in the request sheet”; “this identifier refers to the same material”; “this result is surprising because the earlier state was different.”

The spoken channel carries information that action-only traces cannot. It can identify which visible values are causal,

distinguish a rule from an incidental example, and state a constraint before the corresponding case occurs. The screen and action channels, in turn, ground narration in exact entities, values, and state transitions. Neither is a substitute for the other.

The environment is resettable. The task seed defines the records and values encountered by the demonstrator, while the learner sees only the ordinary application surfaces and the resulting lesson. This gives repeated runs the same world without exposing hidden fixture controls or answers.

3.2 The structured lesson

Brackett compiles the interaction into a structured Show-and-Tell bundle. The bundle is designed to be read as one temporal account, not as a directory of unrelated files. Its logical step contains:

- a Playwright-style action with page and tab continuity;
- the user’s spoken commentary aligned to the active interval;
- the screen state associated with the action;

```
await page.click('Stock Projected Qty')
// User: "I check projected stock before deciding."
```

Screen	ERP report after the menu selection
Atty	link: Stock Projected Qty; textbox: Item Code
Time	after request row selection, before decision write-back

Figure 3: One logical lesson step. Action, voice, screen state, accessibility semantics, and time describe the same moment. The representation is conceptually stable even as the production capture system evolves.

- accessibility semantics, including roles, labels, values, and hierarchy; and
- stable identifiers that connect the action, commentary, screenshot, and metadata.

The channels contribute different evidence. Screenshots preserve layout, visual grouping, canvas-rendered content, and transient menus. Accessibility state supplies exact labels and values when pixels are difficult to read. Actions establish what the person selected or changed. Voice explains why. Temporal keys preserve their correspondence, and an agent may inspect only the modalities it needs.

Raw recordings may be retained for auditing the collection, but they are not the canonical learner input. This design removes decoding, frame sampling, and long-video context limits as confounds. The benchmark asks whether an agent understands the lesson after the relevant perceptual and interaction signals have been coherently represented.

3.3 From demonstration to evaluation item

Each released task contains five parts:

1. **Environment state:** the resettable world in which the lesson occurs.
2. **Demonstration:** the human’s actions and narration.
3. **Structured context:** the bundle delivered to learners.
4. **Question set:** private probes whose answers are supported by the lesson.
5. **Grading contract:** normalized answers or explicit required and forbidden propositions.

The open-source release contains the frozen contexts, environments, question schemas, and evaluator. Active answer material can be served separately for a durable leaderboard, while retired forms remain available for audit. The proprietary collection stack is not required to reproduce a published score: evaluation begins from the versioned bundle delivered to every agent.

3.4 Question families

The quiz is not a memory dump of every label that appeared. It samples complementary forms of learning.

Observed facts. These questions ask about non-obvious values, entities, menu selections, or application states that appeared in the lesson. They measure attention and factual retention. A useful fact is specific to the demonstrated episode

and difficult to recover from general product knowledge.

Temporal procedure. These questions ask what happened before or after another event, which observations preceded a decision, or what final state was actually reached. They distinguish an ordered or causal account from an unordered list of actions.

Cross-system semantics. Some lessons join evidence across tabs and applications. Questions probe which supplier part mapped to which internal material, which spreadsheet field populated an ERP record, or which email corresponded to a queue row.

Decision and rationale. These questions identify the evidence that controlled a demonstrated judgment. They are included only when the lesson establishes the relation through narration, contrast, or repeated behavior; a coincidental action is not promoted into a policy.

Transfer. A new case changes the values while preserving a demonstrated rule. The learner must apply the rule rather than repeat an observed answer. These items remain grounded in what the lesson taught and do not ask for unsupported domain expertise.

Limits of the lesson. Some questions ask what the demonstration did *not* establish. For example, opening a permanent-submit confirmation is not evidence that the record was submitted. These questions reward precise abstention and penalize plausible completion of an unfinished episode.

3.5 Evaluation protocol

Every learner receives the same task-scoped system instruction and the same bundle. It may inspect actions, commentary, screenshots, and accessibility state as needed. It receives no environment controls, answer key, neighboring task, or hidden policy implementation. After studying the lesson, it answers the task’s private quiz.

Closed questions use normalized exact answers. Structured open questions are parsed into required claims. Explanatory questions specify statements that must appear and claims that must not appear. A language model may recognize equivalent wording, but the released rubric and evidence establish the answer.

Questions are evaluated independently when contextual leakage among neighboring probes is a concern. Meaning-preserving wordings share one semantic question identifier and one unit of score, so paraphrases cannot increase the weight of one fact. Related demonstration takes share a question-program identifier but retain their own evidence and gold answers. We also retain the full item-by-agent response matrix for paired analysis.

What the score means. A high score indicates that the agent recovered information and judgment supplied by one structured human lesson. It is not an execution-success score, a general product-knowledge score, or proof of autonomy or long-term learning.

Working-snapshot inventory	Count
Tasks	63
Runnable environments	59
Quiz questions	842
Planted or recorded gaps	176
Human-captured demonstrations	13
Evidence-reviewed human lessons	10
Questions in that reviewed core	100

Table 2: The collection is moving toward release. The final paper will replace these values with counts from one immutable benchmark tag.

4 Dataset Composition

SHOWTELLARENA is a growing collection of situated demonstrations rather than a random sample of webpages. The current software testbed is selected for lessons that require a person to combine visible evidence, application state, and judgment. The working snapshot contains 63 scenarios, 842 questions, and 176 deliberately unshown or incompletely resolved cases. These values are centralized and will be replaced by the frozen release manifest.

4.1 Work represented

The current tasks form seven mutually exclusive work families (Table 3). They include ERP, spreadsheets, mail, accounting, fleet, and document intake, alongside common web settings such as source control, commerce, maps, forums, and reference sites. Many lessons cross two or three systems, reflecting how work moves between records rather than remaining inside one page.

We retain both controlled scenarios and organic web tasks. Controlled environments let us vary entities and boundary cases while preserving exact gold state. Organic environments broaden interface diversity. The primary human-demonstration set is reported separately from earlier scripted lessons so improvements in collection quality do not disappear inside one aggregate.

4.2 Complementary demonstrations

For each human-taught scenario, the release design collects three takes from the same teacher. Two takes repeat the same lesson with the same seeded facts. Their wording, timing, and incidental motions may differ naturally, but a semantically matched question should retain the same answer. The third take changes exactly one sealed fact in the reset environment while holding the subject and remaining facts fixed. The same question program is instantiated against all three lessons; questions affected by the intervention receive a different gold answer, while unaffected questions remain invariant.

One question program, three takes

Take	Shown fact	Gold
A	warehouse: Detroit	Detroit
A'	warehouse: Detroit	Detroit
B, one fact changed	warehouse: Toledo	Toledo

Which warehouse did the operator select?

The intervention is recorded in the release manifest but is not announced to the learner. This produces two controls from one lesson family: A/A' tests robustness to a new performance of the same teaching, while A/B tests whether the response follows the demonstrated fact rather than a prior, a question cue, or the answer from another take. All three takes remain in one split and one statistical cluster.

4.3 Difficulty is multidimensional

Long demonstrations are not necessarily difficult, and short ones are not necessarily easy. We annotate factors that make a lesson intrinsically demanding:

- **Evidence span:** how far apart the supporting observations occur.
- **System breadth:** the number of applications and entity bindings.
- **Temporal depth:** the number of ordered events needed for an answer.
- **State dependence:** whether an answer depends on before/after state.
- **Numerical precision:** arithmetic, thresholds, dates, or quantities.
- **Inference burden:** observation, mapping, rule application, or boundary reasoning.
- **Signal burden:** whether voice, pixels, accessibility, or their combination is necessary.

These annotations support stratified reporting; they are not collapsed into an unvalidated claim that one task is universally “twice as hard” as another. Empirical item difficulty is estimated from the eventual response matrix and compared with the structural annotations.

4.4 Question provenance and review

Questions in the human-demonstration core are written against the delivered lesson, not against an idealized task description. Reviewers examine the actions, aligned narration, screen evidence, and accessibility state. A hidden specification can check that an answer is correct in the environment, but cannot make an answer teachable if the demonstration never supplied it.

Each candidate is reviewed for answer support, scope, ambiguity, relevance, and grading precision. Questions that reward common sense or generic product knowledge are rewritten or removed. A no-demonstration audit supplies the question without the lesson to independent respondents; high blind success is evidence that the item does not measure learning from the demonstration. Full-context human review provides

Work family	Tasks	Representative lessons	Knowledge tested after the lesson
Supply chain & procurement	12	demand planning, constrained allocation, RFQ awards, PCN triage, purchase requisitions, substitutions	cross-system identity, stock arithmetic, ranking, deadline eligibility, side effects
Finance & commercial controls	10	carrier variance, invoice matching, credit release, discount review, payment reconciliation	thresholds, precedence, source lineage, exact calculations, approval rationale
People operations	6	candidate rematching, requisition triage, offer follow-up, recruiter scorecards	status semantics, matching criteria, counts, write-back and communication
Service & governance	4	fleet faults, return authorization, inbox triage, policy monitoring	escalation paths, temporal conditions, exception handling, lesson limits
Collaboration & community	10	GitLab issue work and forum moderation	title classification, rule precedence, idempotence, untouched-state constraints
Commerce	11	carts, wishlists, reviews, order triage, price sanity	selection, budgets, dates, stock state, competing business rules
Maps & knowledge navigation	10	routing, nearby amenities, Wikipedia lookup and multi-hop comparison	source selection, tie breaking, units, missing evidence, multi-hop binding

Table 3: The working task map. Counts will move before launch. The collection is designed to test different kinds of knowledge, not simply more websites.

Channel	What it establishes best
Actions	selection, input, navigation, and ordering
Voice	intent, decision rationale, and salience
Screens	layout, visual grouping, transient state, canvas content
Accessibility	exact labels, roles, values, and hierarchy
Environment	reset state and reproducible outcomes

Table 4: Complementary lesson channels. No channel is universally privileged. The aligned bundle preserves their agreement and disagreement.

the opposite control: a difficult question that attentive humans cannot answer is not retained merely because models fail it.

The release reports the complete funnel from candidate to accepted item. This prevents a large generation count from being confused with a large valid benchmark.

5 Quality and Release Controls

A demonstration benchmark can fail quietly. A screenshot may omit the value named by a question; voice transcription may distort an identifier; an accessibility snapshot may contain an element that was never visible; a replay may reach a different state; or a plausible answer may come from prior knowledge. We therefore verify the lesson context and each question separately.

5.1 One lesson, several sensors

Pixels, accessibility state, action traces, and speech are correlated but imperfect sensors. Their roles are complementary:

The benchmark does not equate presence in an accessibility tree with human exposure. This is not a hypothetical con-

cern: ScreenQA documents occluded, ghosted, and out-of-sync Android view hierarchies, and 48% of its answer annotations used manually drawn boxes rather than hierarchy leaves alone [Hsiao et al. \(2025\)](#). A hidden menu item, stale node, or off-screen value supports a question only if the lesson representation establishes that it was available to the learner under the evaluated access policy. Conversely, a value visibly rendered in a spreadsheet canvas remains legitimate evidence even when no accessibility node contains it.

Exact speech strings are handled cautiously. Natural narration is valuable precisely because it is not a dictated script, and automatic transcription can be noisy. Voice supports intent and relationships; exact identifiers and quantities should be corroborated by clear screen or accessibility evidence when available.

5.2 Context fidelity

For each released lesson, stable keys bind the temporal step, action, commentary, screenshot, and accessibility state. The release manifest hashes these artifacts and records the context-construction version. A learner’s input is therefore reproducible even if the production capture stack evolves after the release.

Where a demonstration is replayed against a resettable environment, semantic checkpoints verify the screen identity, acted element, important labels, and task-owned state. Pixel identity is neither required nor sufficient: responsive layout may move an element without changing the lesson, while a visually similar replica may omit the exact field or menu structure that carried the teaching signal.

5.3 Question validity

An accepted question must pass four independent checks:

(a) People: requisition triage.

(b) Supply: constrained allocation.

(c) Finance: payment reconciliation.

(d) Service: fleet fault codes.

(e) Documents: invoice intake.

(f) Coordination: confirmation mail.

Figure 4: One benchmark, several kinds of work. Examples are drawn from the current collection rather than one legacy example. The visual state ranges from dense spreadsheets and transactional systems to fleet boards, document queues, and email; the associated questions test facts, mappings, decisions, transfer, and what the lesson did not establish.

1. **Evidence closure:** every required answer claim is supported by the delivered lesson.
2. **Unique grading:** equivalent correct forms are accepted, while materially different claims remain distinguishable.
3. **Demonstration dependence:** respondents without the lesson cannot answer reliably from the question, scenario, or answer options alone.
4. **Attentive answerability:** independent humans with the complete lesson can answer and identify the supporting evidence.

Question writers do not grade their own items. Disagreements about objective facts are adjudicated and retained in the audit record. Usefulness is reviewed separately: two reviewers can agree that a label appeared while disagreeing about whether it is worth testing.

5.4 Leakage and contamination

Questions derived from one lesson share entities, vocabulary, and evidence. They cannot be treated as independent train and test examples. Dataset partitions keep a complete demonstration family, its alternate takes, scenario seed, questions, and paraphrases on one side. Publicly released active answers are auditable but not secure; a long-lived hosted evaluation should serve sealed questions and rotate scenario values, publishing retired forms later.

Within one evaluation, questions are isolated or grouped conservatively so a neighboring prompt does not reveal terminology needed by another. Multiple wordings of the same semantic question share one component identifier and one unit of score.

5.5 Open evaluation boundary

The open release includes the material needed to reproduce an agent score: frozen lesson contexts, environment definitions, questions, grading contracts, and aggregation logic. It also publishes per-question outcomes and the exact release identity used by the paper.

Brackett’s context-construction models and production stack are not part of the benchmark release. This limits reproduction of collection, but not evaluation over the released dataset. We make that boundary explicit rather than describing the proprietary pipeline as an open method. Future independently collected lessons can enter only after they satisfy the same public context and question contracts.

6 Agent Evaluation

Our experiments are designed to answer three questions. First, how much of a human lesson can current agents recover from the full structured context? Second, which parts of the context make that learning possible? Third, does an answer follow the demonstrated evidence, or would the agent give it when the lesson stays the same or changes?

6.1 Learners and controlled input

We evaluate Brackett and representative frontier agents from Anthropic and OpenAI. Exact product, model, version, inference settings, and run date are frozen in the release manifest. The primary comparison is context-controlled: every learner receives identical system instructions and the same Show-and-Tell bundle. This isolates the learner from differences among native recording interfaces.

An optional end-to-end condition evaluates complete commercial teach experiences. We report it separately because the products may capture different modalities and create different internal artifacts. Combining these two settings into one leaderboard would obscure whether a failure arose in collection or comprehension.

Each agent studies one lesson in a fresh session and answers questions without access to the source environment. Runs are repeated under a balanced schedule. Invalid output, timeout, refusal, and infrastructure failure remain distinct outcomes; zero scores are never silently excluded.

6.2 Metrics

Question accuracy. Closed and structured questions receive binary correctness after normalization. Explanatory questions receive a bounded score over required and unsupported claims. The headline score macro-averages distinct semantic questions.

Family accuracy. We report observation, temporal, cross-system, decision, transfer, and lesson-limit questions separately. This reveals whether an agent remembered the episode but failed to generalize, or generalized while inventing facts that were never demonstrated.

Paired discordance. For every pair of agents we retain all four cells: both correct, only agent A correct, only agent B correct, and neither correct. The off-diagonal cells are the evidence that the benchmark distinguishes the agents. Aggregate accuracy without discordance may describe a saturated or uniformly impossible test.

Take consistency and sensitivity. For the same agent, we also pair semantically matched questions across complementary takes. When two takes teach the same facts, we measure semantic answer agreement and paired correctness. When one sealed fact changes, we measure whether both answers match their take-specific gold values and whether the agent incorrectly repeats an unchanged answer.

Grounding and abstention. For questions that require evidence references or explicit uncertainty, we score whether the response stays within the lesson. A plausible answer that exceeds the demonstrated evidence is not treated as equivalent to a grounded answer.

Draft status. The dataset is expanding and the final balanced comparison has not been frozen. We preserve the result slots rather than copy stale pilot numbers into a new scientific claim. The camera-ready artifact will name coverage, repeats, failures, uncertainty, and the exact benchmark release.

6.3 Context ablations

The full bundle is compared with controlled views of the same lesson:

- **Question only:** measures blind guessability and prior knowledge.

- **Voice only:** tests how much of the taught content narration preserves alone.
- **Actions + voice:** removes screen and accessibility evidence.
- **No voice:** retains actions, screens, and accessibility to measure the value of explanation.
- **No screenshots:** measures the contribution of pixels beyond actions and accessibility.
- **No accessibility:** measures exact semantic structure beyond pixels.
- **Shuffled time:** preserves local observations while breaking procedure and causal adjacency.

These are evaluated by question family. We expect voice removal to matter most for decision rationale and transfer, screenshots for visual grouping and canvas content, accessibility for exact labels and values, and temporal shuffling for procedure. The experiment, rather than the representation’s design intent, determines whether those expectations hold.

6.4 Demonstration and wording ablations

The three-take design provides a second family of ablations. Each agent sees one take at a time in a fresh session. Takes A and A’ teach the same lesson and facts through independent performances by the same teacher; their semantically matched answers should agree even when narration and timing differ. Take B changes one sealed fact. A strong learner should keep answers to unaffected questions fixed and change only answers supported by that intervention.

Meaning-preserving question paraphrases provide another axis. They are rotated across repeated runs rather than shown together, and all wordings retain one semantic question identifier. We expect the same semantic answer and comparable accuracy distributions, not identical stochastic samples.

Speech realization provides two further ablations. The original three takes retain the teacher’s human voice. For selected lessons, we then render the verified transcript with text-to-audio (T2A) speakers that vary voice and accent while preserving the words. These variants pass through the same lesson-construction path, so the semantic answer should remain fixed. Selected task rows are also repeated by different people. This changes not only accent, but natural phrasing, emphasis, ordering, and what the teacher chooses to point out. We therefore report it as cross-teacher robustness rather than a pure acoustic ablation, and retain a matched question only when every delivered lesson independently supports its answer.

A secondary contrast condition gives the learner A and B together and asks which fact changed and which answers should therefore differ. This directly tests comparison across lessons, but it is reported separately because the primary benchmark gives the learner only one demonstration.

The take triad can also be crossed with selected context ablations. For example, if the Detroit/Toledo intervention is visible on screen but never spoken, counterfactual sensitivity should fall when screenshots are removed but remain under a no-voice condition. We run the full context on every triad and preregister targeted crosses where one modality carries the changed fact; an exhaustive Cartesian product would spend

Learner	Overall	Observed	Temporal	Cross-system	Decision	Transfer	Boundary
Brackett agent	–	–	–	–	–	–	–
Claude	–	–	–	–	–	–	–
Codex	–	–	–	–	–	–	–
Human, full lesson	–	–	–	–	–	–	–

Table 5: Release evaluation table. Dashes are intentional in this working manuscript. Final values will be populated only from the balanced, frozen run matrix associated with the submitted dataset release.

Condition	Overall	Decision	Temporal
Full structured lesson	–	–	–
No voice	–	–	–
No screenshots	–	–	–
No accessibility	–	–	–
Shuffled time	–	–	–
Question only	–	–	–

Table 6: Context ablations. The final table will report paired changes on the same semantic question components, rather than scores from unrelated item subsets.

Paired-ablation metric	Result
Same-fact A/A' semantic agreement	–
Same-fact A/A' both-correct rate	–
One-fact A/B both-correct rate	–
One-fact A/B unchanged-on-flip error	–
Question-wording semantic agreement	–
Same-transcript T2A semantic agreement	–
Cross-teacher semantic agreement	–
Cross-teacher both-correct rate	–
Joint A+B difference-question accuracy	–

Table 7: Demonstration and wording ablations. Cells are intentionally blank until the balanced run is frozen. The final paper will expand each paired rate into both/first-only/second-only/neither cells and visualize them as heatmaps. Joint A+B is separate because it gives the learner two lessons.

runs without adding an identifiable contrast.

These combinations multiply controlled observations, not independent questions. The question program remains the unit of macro-averaging, and uncertainty is clustered by demonstration family across takes, wordings, voices, and teachers. Agent–agent discordance and within-agent ablation discordance are reported separately: the first compares two learners on the same lesson, while the second asks whether one learner remains stable across equivalent questions and teachings, and follows the changed evidence across A/B. An unchanged answer on a changed-gold pair is marked separately from an ordinary wrong answer because it directly indicates anchoring, copying, or failure to use the demonstration.

6.5 How we will read the results

A useful benchmark should reveal different failure profiles. We therefore examine item-level clusters rather than declaring one universal winner from a small mean difference. Examples include agents that:

- recover exact values but lose cross-tab entity identity;
- reproduce the observed decision but cannot state the controlling evidence;
- infer a sensible generic rule that contradicts the demonstrated procedure;
- over-complete an unfinished action instead of reporting what the lesson actually reached; or
- use narration effectively but fail on information exposed only in a transient screen state.

For paired comparisons we report bootstrap intervals clustered by demonstration family, not by take or wording. We also publish the item-by-agent response matrix, allowing later work to estimate difficulty and discrimination without treating correlated questions from the same lesson as independent.

7 Limitations and Conclusion

What is measured. SHOWTELLARENA measures immediate comprehension after one situated lesson. It does not yet prove that an agent can retain the knowledge over time, act on it reliably, recover from environmental changes, or know when to request additional teaching. Comprehension is a necessary layer between demonstration and learning, not the complete problem.

Collection boundary. The released contexts are produced by Brackett’s proprietary Show-and-Tell stack. This makes the evaluation reproducible from frozen bundles while leaving independent reproduction of the collection pipeline incomplete. We disclose that boundary and release the full learner input rather than presenting a proprietary transformation as an open method.

Dataset scope. The lessons are authored and selected, not sampled uniformly from everything people might teach. The collection currently emphasizes browser-based enterprise and web scenarios with English narration. Results should be stratified by scenario and question family, not generalized to every kind of human teaching.

Question judgment. Question usefulness and explanatory grading require human judgment. Blind audits, evidence review, explicit rubrics, and released grading traces reduce but do not eliminate subjectivity. Active questions also require rotation to limit contamination.

We have presented SHOWTELLARENA, an evaluation of whether agents can acquire situated knowledge from one human demonstration. Each agent receives the same structured lesson: actions, spoken reasoning, screen states, accessibility

ity semantics, and temporal continuity. Questions then test the facts, mappings, decisions, transferable rules, and limits learned from that lesson. This turns “show it once” into a reproducible comparison and measures which parts of a human demonstration survive in an agent’s understanding.

Each lesson also becomes a Knowledge Graph whose entities, claims, and dependencies remain tied to their source moments. Verified facts from this compounding graph can support grounded answers, evaluations, and training. This paper focuses on the benchmark: learning from frozen lessons.

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A Qualitative Examples

Figure 4 gives the main-paper overview. Figure 5 enlarges four cases from the current collection. Each teaches different situated knowledge. Requisition triage requires status aggregation; limited inventory requires precedence and arithmetic; an RFQ award combines delivery eligibility with total cost; and a fleet fault combines vehicle state with code meaning.

B Canonical Context Card

For audit and qualitative analysis, a logical step can be rendered as a context card:

Step 17: open the stock report

Action `click('Stock Projected Qty')`
Commentary “I check projected stock for the requested item before I decide.”
Visible state Stock workspace with the report menu expanded
Accessibility link: Stock Projected Qty;
textbox: Item Code
Temporal links after request-row selection; before item lookup and writeback
Observer access action, commentary, screenshot, and ally available

The learner receives the underlying artifacts rather than this condensed card. The card illustrates the alignment contract and gives reviewers a compact way to inspect a question’s evidence.

C Question and Grading Contracts

Closed answer. A question defines a canonical value and normalized aliases. Normalization handles formatting differences, not semantic disagreement. For example, an internal identifier may accept case variation but not a neighboring part number.

Structured answer. Mapping, sequence, and comparison questions define typed slots. A response receives credit only when the required entities occupy the correct roles; mentioning both identifiers without stating their direction is insufficient.

Explanatory answer. Explanatory answers use explicit positive and negative propositions. A terminal-state question may require *draft record exists* and *confirmation dialog is visible*, while forbidding the unsupported claim *record was submitted*. This lets a grader distinguish omission from fabrication.

Unanswerable question. A lesson-limit item defines why the available evidence is insufficient. Generic uncertainty language is not enough if the answer also asserts an unsupported value.

D Full Reporting Contract

Every published result should identify:

- benchmark version, manifest digest, and release date;
- included, blocked, failed, and excluded demonstrations;
- exact learner product/model versions and inference settings;
- number of repeats and the run schedule;
- demonstration-family and take identifiers, including the teacher linkage;
- the sealed intervention, affected question programs, and invariant controls;
- context condition and modality availability;
- question-program, wording, and grading-contract digests;
- raw response retention and judge configuration;
- zero, timeout, refusal, and infrastructure-failure policy;
- aggregation unit and cluster used for uncertainty; and
- the complete paired discordance table.

	Agent B correct	Agent B wrong
Agent A correct	both	A only
Agent A wrong	B only	neither

Table 9: Agent-agent paired comparison. A-only and B-only questions are the effective comparative sample. All four cells remain visible; take consistency is reported in a separate matrix with the same four-cell accounting.

E Native Teach Interfaces

The primary paper comparison controls the learner input. SHOWTELLARENA can also evaluate complete teach experiences, in which each product captures and summarizes the demonstration through its own interface. These results answer a broader product question and are therefore reported separately.

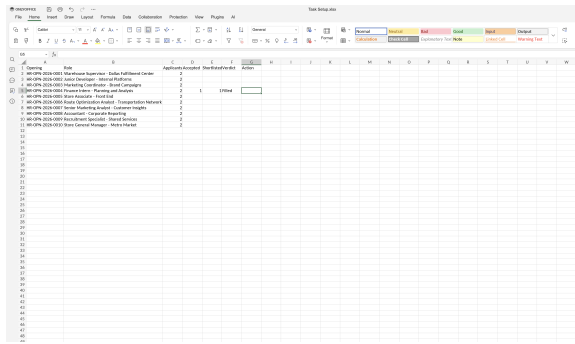
F Dataset Card Summary

Intended use. Measure immediate learning from one structured human demonstration, compare learner failure profiles, and study which context channels support comprehension.

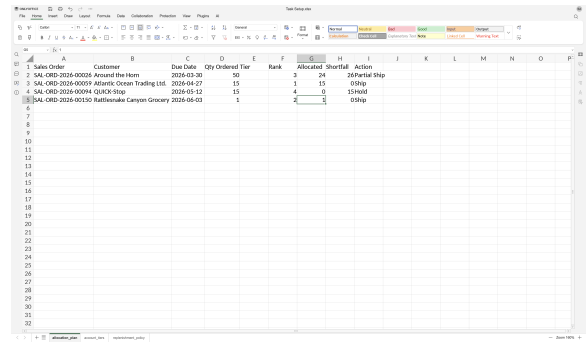
Out of scope. The dataset is not an execution benchmark, a certification of autonomous-system safety, or a complete sample of what people teach through software. It should not be used to infer workforce capability or evaluate individual demonstrators.

Privacy. Released demonstrations use task-owned or consented content and must be reviewed for personal information, credentials, and third-party confidential data. Secrets used to access an environment are not part of the learner context.

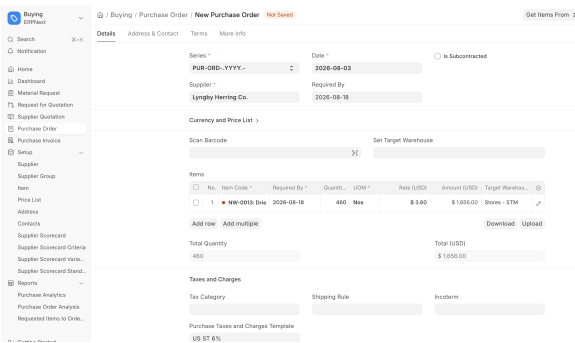
Maintenance. Active evaluation forms may rotate. A paper result remains bound to its immutable release, while later releases can add demonstrations, repair questions, and improve coverage without changing the historical score.



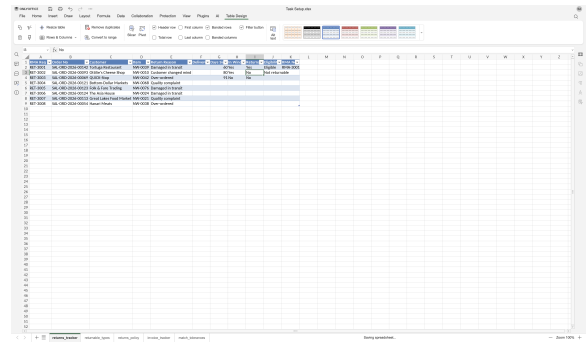
(a) People operations: applicant evidence becomes a requisition verdict and follow-up action.



(b) Inventory allocation: tier and due date control rank, allocation, shortfall, and action.



(c) Procurement: a purchase order records the quote that survives delivery and price checks.

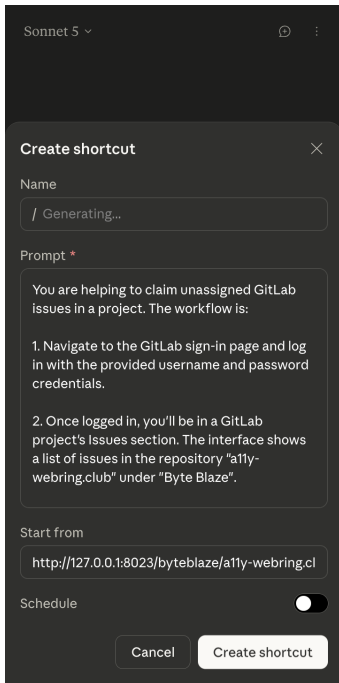


(d) Returns: delivery date and item-group policy determine eligibility and the RMA side effect.

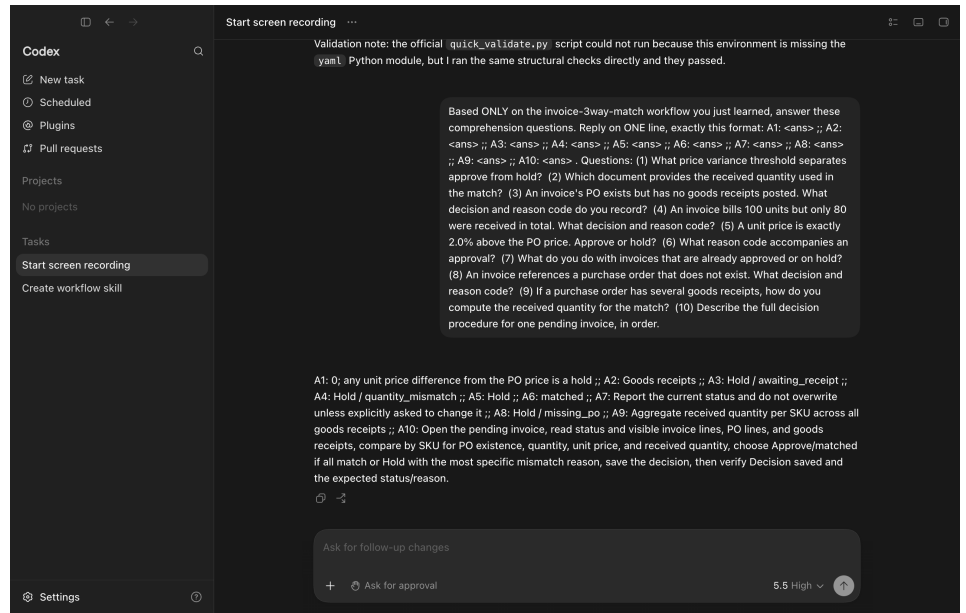
Figure 5: Larger examples from four work families. The benchmark asks what an attentive learner can recover from the demonstrated relationships, not whether it recognizes that a spreadsheet, ERP form, or table is on screen.

Case	Lesson-specific evidence	Representative question form
Requisition triage	applicant statuses, accepted and shortlisted counts, tracker writeback	Which statuses counted as progress, and how did the resulting counts map to verdict and action?
Limited-stock allocation	ERP stock tuple, account-tier notes, due dates, ranked worksheet	In what order were customers ranked, and what allocation, shortfall, and action remained for each?
RFQ quote award	need-by date, quote arrival, quantity, rate, total, created PO	Why did a lower-priced quote lose, which quote won, and where was the resulting PO bound?
Payment reconciliation	processor payment detail, intent identifier, gross and fee values, clearing account	Which identifier and amount crossed systems, and which visible values were not entered as the deposit?
Fleet fault triage	vehicle shop state, SPN/FMI meaning, service history, board order	Which vehicle could dispatch, which required a stop, and what evidence controlled the difference?
Return authorization	policy sheet, delivery date, item group, tracker state, RMA side effect	Which external facts established eligibility, and did the demonstration show an RMA being created?

Table 8: Examples are organized by taught relationship, not by website. Exact active wording and answers remain release-bound; the forms show why each item requires the demonstration.



(a) A generated teach shortcut.



(b) A recorded lesson followed by comprehension questions.

Figure 6: End-to-end product evaluation is a secondary condition. Native products create different internal artifacts and may capture different modalities. The context-controlled benchmark isolates the learner; this condition evaluates the complete experience.